

PES UNIVERSITY

Department of Computer Science & Engineering

Lab 2: Agile-Scrum with Jira

Problem Statement #05 — Academic Elective Bidding &
Allocation System

Course: Software Engineering Lab

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Methodology: Agile-Scrum (2 Sprints, 1-week each)

OVERVIEW

Project Summary

System: Academic Elective Bidding & Allocation System

Actors: Student, Academic Registrar, Instructor

Context: University departments need an automated elective course selection mechanism where students allocate bidding credits across preferences and the engine solves allocation constraints while preventing timetable collisions.

3

EPICS

12

USER STORIES

2

SPRINTS

55TOTAL STORY
POINTS

DELIVERABLE 1

EPICs & User Stories

EPIC 1 — Student Bidding & Credit Allocation

21 Story Points

Enable students to browse elective courses, distribute 100 bidding credits across ranked preferences, validate prerequisites, and submit bids within the configured bidding window.

ID	USER STORY	SP	PRIORITY
US-001	As a Student , I want to view all available electives with course details, instructor, schedule, and seat count so that I can make informed bidding decisions.	3	High
US-002	As a Student , I want to distribute 100 bidding credits across my ranked elective preferences so that I can indicate the strength of my choice.	5	High
US-003	As a Student , I want the system to validate my prerequisite completions before accepting a bid so that I don't bid on courses I'm ineligible for.	5	High
US-004	As a Student , I want the system to detect timetable collisions among my selected courses so that I avoid schedule conflicts.	5	High
US-005	As a Student , I want to save my bid as a draft and finalize it later within the bidding window so that I can review before submitting.	3	Medium

EPIC 2 — Allocation Engine & Registrar Administration

21 Story Points

Allow the Academic Registrar to configure bidding parameters, run the constraint-solving allocation engine on collected bids, and manage the overall bidding lifecycle.

ID	USER STORY	SP	PRIORITY
US-006	As the Registrar , I want to configure bidding window dates, per-course seat caps, and max credits per course so that the cycle matches institutional needs.	3	High
US-007	As the Registrar , I want to trigger the allocation engine to process all submitted bids and produce seat assignments that respect capacity and credit rankings.	8	High
US-008	As the Registrar , I want to view a dashboard showing bid statistics (total bids per course, average credits, fill rates) so I can monitor demand.	5	Medium
US-009	As the Registrar , I want to manually override allocations for special cases (e.g., final-year students) so that edge cases are handled fairly.	5	Low

EPIC 3 — Results & Notifications

13 Story Points

Publish allocation results to students and instructors, send notifications, and allow stakeholders to view and export their final course assignments.

ID	USER STORY	SP	PRIORITY
US-010	As a Student , I want to view my allocation results showing which courses I was assigned so that I can plan my semester.	3	High
US-011	As an Instructor , I want to view the final student roster for my elective course so that I can prepare course materials accordingly.	3	Medium
US-012	As a Student , I want to receive an email/notification when allocation results are published so that I'm promptly informed.	2	Medium

EPICs Summary

EPIC	NAME	USER STORIES	TOTAL STORY POINTS	PRIORITY BREAKDOWN
EPIC 1	Student Bidding & Credit Allocation	US-001 to US-005	21	4 High, 1 Medium
EPIC 2	Allocation Engine & Registrar Admin	US-006 to US-009	21	2 High, 1 Medium, 1 Low
EPIC 3	Results & Notifications	US-010 to US-012	13	1 High, 2 Medium
	Total	12 Stories	55	7 High, 4 Medium, 1 Low

SPRINT PLANNING

Sprint Breakdown

Sprint 1 — Foundation (Week 1)

Goal: Build core student bidding flow and registrar configuration. **Capacity:** 29 Story Points

STORY	TITLE	SP	PRIORITY	STATUS
US-001	View available electives catalogue	3	High	Done
US-002	Distribute 100 bidding credits	5	High	Done
US-003	Validate prerequisite completions	5	High	Done
US-004	Detect timetable collisions	5	High	Done
US-006	Configure bidding parameters	3	High	Done
US-007	Run allocation engine	8	High	Done
Sprint 1 Total		29		Done

Sprint 1 — Final Scrum Board State

To Do (0)

All items completed

In Progress (0)

All items completed

Done (6)

US-001View electives3 SP

US-002Distribute credits5 SP

US-003Validate prereqs5 SP

US-004Timetable collisions5 SP

US-006Bidding params3 SP

US-007Allocation engine8 SP

Sprint 2 — Enhancement & Delivery (Week 2)

Goal: Complete remaining features — results, notifications, admin dashboard, and drafts. **Capacity:** 26 Story Points

STORY	TITLE	SP	PRIORITY	STATUS
US-005	Save bid as draft	3	Medium	Done
US-008	Registrar bid statistics dashboard	5	Medium	Done
US-009	Manual allocation override	5	Low	Done
US-010	Student views allocation results	3	High	Done
US-011	Instructor views student roster	3	Medium	Done
US-012	Allocation result notification	2	Medium	Done
	Sprint 2 Total	21		Done

Sprint 2 — Final Scrum Board State

To Do (0)

All items completed

In Progress (0)

All items completed

Done (6)

US-005

Save draft bid

3 SP

US-008

Bid statistics

5 SP

US-009

Manual override

5 SP

US-010

View results

3 SP

US-011

Instructor roster

3 SP

US-012

Notifications

2 SP

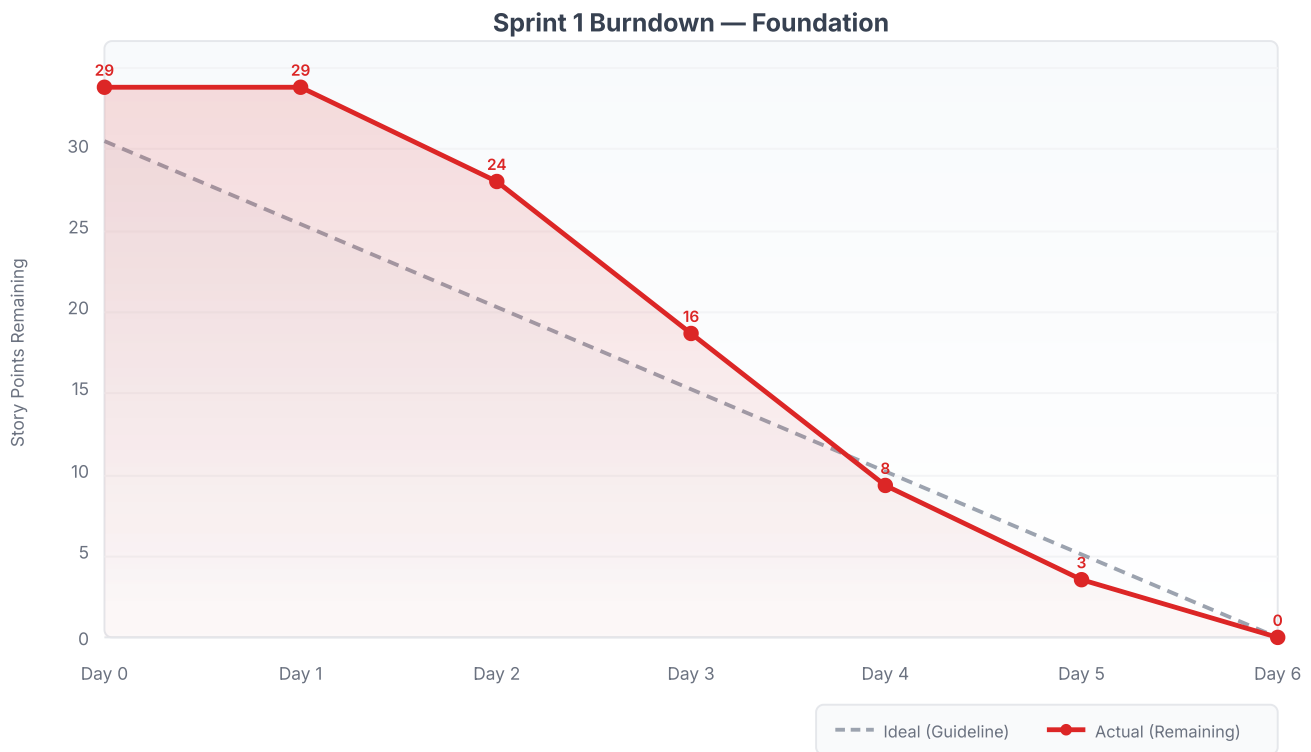
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DELIVERABLE 2

Burndown Chart

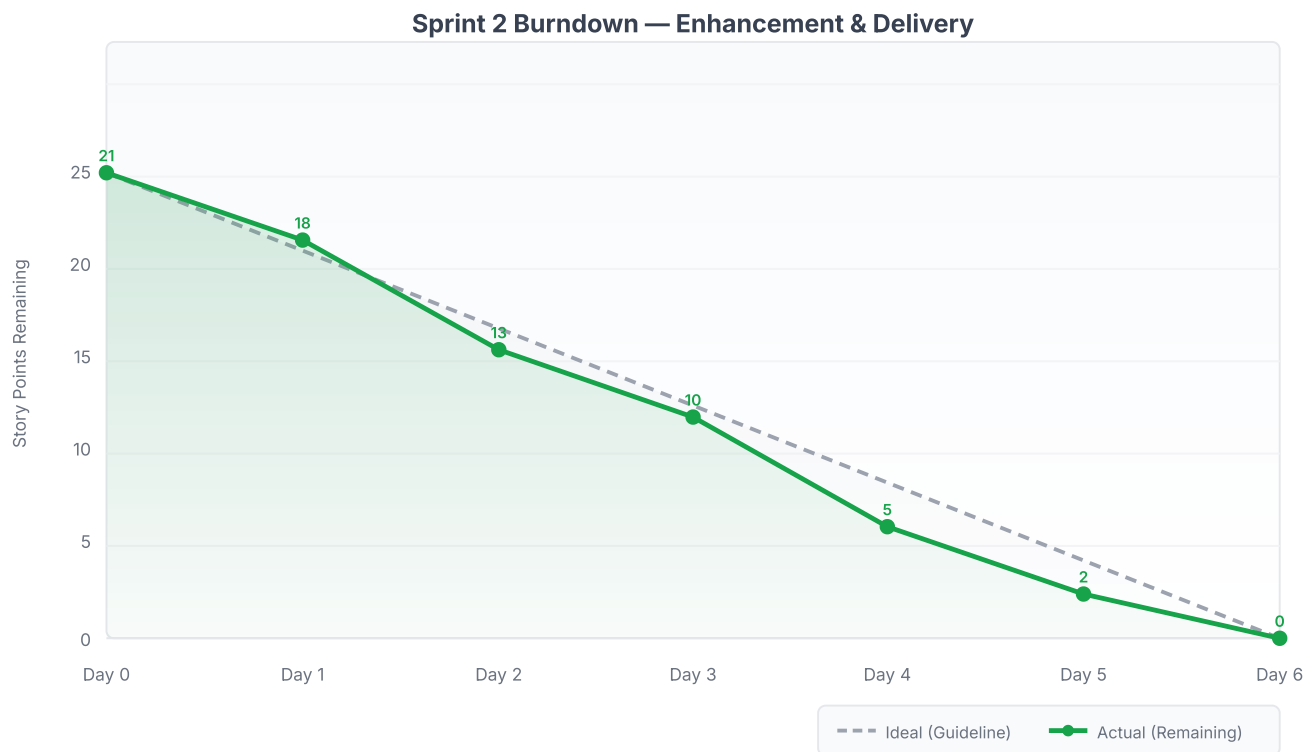
Sprint 1 — Burndown Chart (29 Story Points over 7 Days)



Sprint 1 Observations

- **Day 1:** Planning and environment setup — no stories completed (29 SP remaining).
- **Day 2:** Completed US-001 (View Electives, 3 SP) + US-002 partially. Remaining: 24 SP.
- **Day 3:** Completed US-002 (5 SP) + US-003 (5 SP, Prerequisite Validation) — engine heavy. Remaining: 16 SP.
- **Day 4:** Completed US-004 (Timetable Collisions, 5 SP) + US-006 (Config, 3 SP). Remaining: 8 SP.
- **Day 5:** US-007 (Allocation Engine, 8 SP) mostly complete, 3 SP remaining for integration testing.
- **Day 6:** All stories completed. Sprint closed successfully with **0 SP remaining**.

Sprint 2 — Burndown Chart (21 Story Points over 7 Days)



Sprint 2 Observations

- **Day 1:** Completed US-005 (Save Draft, 3 SP). Remaining: 18 SP.
- **Day 2:** Completed US-008 (Bid Statistics Dashboard, 5 SP). Remaining: 13 SP.
- **Day 3:** Completed US-011 (Instructor Roster, 3 SP). Remaining: 10 SP.
- **Day 4:** Completed US-009 (Manual Override, 5 SP). Remaining: 5 SP.
- **Day 5:** Completed US-010 (View Results, 3 SP). Remaining: 2 SP.
- **Day 6:** Completed US-012 (Notifications, 2 SP). Sprint closed with **0 SP remaining**.

REFLECTION

Reflection Questions

1. Did your estimations reflect the actual effort?

Overall, the story point estimates were reasonably accurate. The Allocation Engine (US-007, 8 SP) was correctly identified as the most complex item — it consumed nearly 2 full days of sprint capacity. Smaller stories like View Electives (3 SP) were completed quickly as expected. The Fibonacci scale helped distinguish between trivially simple tasks and genuinely complex ones. The Sprint 1 burndown shows a flat line on Day 1 due to planning overhead, which could have been anticipated better.

2. Was your backlog well-prioritized?

Yes. All **High-priority** stories were assigned to Sprint 1, ensuring core bidding and allocation functionality was delivered first. Sprint 2 focused on Medium and Low priority enhancements (dashboards, notifications, draft saving). This MoSCoW-aligned approach ensured that even if Sprint 2 was cut short, the system's essential workflow (bid → allocate → view results) would be complete.

3. How did your simulated sprint align with your plan?

Both sprints completed all planned stories with 0 SP remaining, aligning well with the plan. Sprint 1 had a slightly delayed start (Day 1 plateau) but caught up by Day 3 through accelerated development. Sprint 2 tracked very closely to the ideal guideline, suggesting better estimation maturity from the experience gained in Sprint 1. The team velocity stabilized around 25 SP per sprint.

4. What insights did the burndown chart give about your team's capacity?

Sprint 1: The initial flat period (Day 0–1) indicates setup overhead. The steep descent from Day 2–4 shows the team hit its stride once the architecture was in place. The chart confirms a velocity of ~29 SP/week.

Sprint 2: A smoother, more linear burndown indicates consistent delivery — the team had stabilised its workflow. The actual line tracked closer to the guideline, suggesting improved estimation accuracy. Both charts confirm no scope creep occurred, which is ideal for a focused project.